Module 03 — Object-Oriented Programming (OOP)

Phase: Advanced Python

Previous: Modules and Packages
Next: Exception Handling (after Part 4)

Module Outcome

Model application entities as classes: bundle data with behaviour, specialise with inheritance, share one call site across types, and lock incomplete blueprints with ABC.

Work the four parts **in order**. Do not skip ahead.

Part	File	Focus
1	03_OOP_Part1_Classes_Encapsulation.md	Dict pain → class / `self` / `__init__`, variables, methods, encapsulation, BankAccount warm-up
2	03_OOP_Part2_Inheritance.md	IS-A, inheritance types, override, `super()`, HAS-A composition
3	03_OOP_Part3_Polymorphism_Project.md	Polymorphism, duck typing, `__str__`, BankAccount CLI
4	03_OOP_Part4_Abstract_Classes_Interfaces.md	Abstract methods, `ABC`, interface / Protocol, Campus Wallet

Lab kits: 15_Lab_Assets/17_bank_account_oop/ (Part 1 warm-up → Part 2 `SavingsAccount` → Part 3 CLI) · 15_Lab_Assets/32_campus_wallet_abc/ (Part 4)

Course order: Exception Handling comes **after** all four OOP parts. OOP examples and labs validate with **`print` and `return`**. Do not require `try` / `except` / `raise` here.

Optional enrichment: Professional Python Deep Dive (composition patterns, `__repr__` — ABC is taught in Part 4)